Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		to ensure that the concentration of $AgNO_3$ in the burette is accurate / does not change			1	1		1
		(ii)		award (1) for any of following • undiluted seawater would give too high a titre / a titre of 132 cm^3 • concentration of Cl^- ions is too high (in undiluted seawater) • reduces Cl^- ion concentration (so less $AgNO_3$ needed)			1	1		1
		(iii)		use a pipette/burette to measure seawater (1) into a volumetric flask (1) add 50 cm^3 of seawater to a 250 cm^3 volumetric flask (and make up to the mark with deionised water) / 20 cm^3 of seawater to a 100 cm^3 volumetric flask (and make up to the mark with deionised water) (1)	1 1		1	3	1	3
		(iv)		award (1) for any action and (1) for corresponding explanation e.g. add drop by drop … to avoid 'overshooting' the endpoint shake the flask/rinse the side of the flask … to ensure that all the reactants react place flask on a white tile … to clearly see the colour change			2	2		2
		(v)		$2Ag^+(aq) + CrO_4^{2-}(aq) \rightarrow Ag_2CrO_4(s)$ ignore state symbols		1		1		

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(vi)		moles Ag^+ = $\frac{0.100 \times 26.40}{1000}$ = 2.64×10^{-3} (1) moles Cl^- in 25.0 cm^3 diluted sample = 2.64×10^{-3} moles Cl^- in 25.0 cm^3 undiluted sample = $2.64 \times 10^{-3} \times 5 = 0.0132$ (1) moles Cl^- in 1000 cm^3 undiluted sample = $0.0132 \times 40 = 0.528$ (1) mass Cl^- in 1000 cm^3 seawater = $0.528 \times 35.5 = 18.7$ g (1) answer **must** be to 3 sig figs		4		4	3	
	(b)	(i)		pour <u>seawater</u> into a beaker and <u>add barium nitrate solution</u> (and allow precipitate to settle) (1) add more drops of barium nitrate (to the clear solution) until no further precipitate forms (1) filter precipitate (making sure that all of it is transferred to the filter) (1) wash precipitate and dry (1) award (1) for either of following • heat to constant mass • subtract mass of filter paper from mass of precipitate and filter paper	5			5		5
		(ii)		moles $BaSO_4$ = $\frac{0.65}{233.1}$ = 2.789×10^{-3} (1) moles $Ba(NO_3)_2$ = 2.789×10^{-3} volume $Ba(NO_3)_2$ = $\frac{2.789 \times 10^{-3}}{0.100}$ = 0.02789 dm^3 (1) volume = 27.89 / 27.9 / 28 cm^3 (1)		3		3	2	

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
		(iii)		award (1) for any of following • concentrations of barium nitrate solution and sulfate ions in seawater are different • concentration of barium nitrate solution is higher than that of sulfate ions in seawater • concentration of sulfate ions in seawater is lower than that of barium nitrate solution			1	1		
				Question 11 total	**7**	**8**	**6**	**21**	**6**	**12**